



# Sub-chronic exposure of *Oreochromis niloticus* to environmentally relevant concentrations of smaller microplastics: Accumulation and toxico-physiological responses

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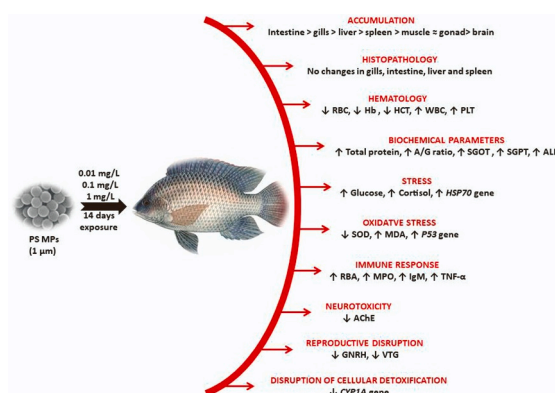
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## HIGHLIGHTS

- 14 days PS MPs exposure on tilapia at environmentally relevant concentration S.
- PS MPs accumulation in the intestine, gills, liver, spleen, muscle, gonad and brain.
- No histopathological alterations in the gills, intestine, liver and spleen.
- Altered hematological, serum biochemical, immune and stress parameters.
- Failure of oxidative, nervous, reproductive and cellular detoxification systems.

## GRAPHICAL ABSTRACT



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## ABSTRACT

This study assesses the accumulation and toxic effects of environmentally relevant concentrations (0.01, 0.1 and 1 mg/L) of polystyrene MPs (1 µm) in *Oreochromis niloticus* (Nile tilapia) for 14 days. The results showed that 1 µm PS-MPs accumulated in the intestine, gills, liver, spleen, muscle, gonad and brain. RBC, Hb and HCT showed a significant decline, while WBC and PLT showed a significant increase after the exposure. Glucose, total protein, A/G ratio, SGOT, SGPT and ALP showed significant increments in 0.1 and 1 mg/L of PS-MPs treated groups. The increase in cortisol level and upregulation of *HSP70* gene expression in response to MPs exposure indicate MPs-mediated stress in tilapia. MPs-induced oxidative stress is evident from reduced SOD activity, increased MDA levels and upregulated *P53* gene expression. The immune response was enhanced by inducing respiratory burst activity, MPO activity and serum TNF-α and IgM levels. MPs exposure also led to down-regulation of *CYP1A* gene and decreased AChE activity, GNRH and vitellogenin levels, indicating the toxicity of MPs on the cellular detoxification mechanism, nervous and reproductive systems. The present study highlights the tissue

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accumulation of PS-MP and its effects on hematological, biochemical, immunological and physiological responses in tilapia with low environmentally relevant concentrations.

## 1. Introduction

The plastic particles with a size ranging from 1  $\mu\text{m}$  to 5 mm are categorised as microplastics (MPs) [32]. These ubiquitous pollutants are present in oceans, freshwater and terrestrial ecosystems and lead to accumulation and detrimental effects on biota [107,27,28,48,5]. Recently, Garcés-Ordóñez et al. [35] reported that the mean abundances of MPs in water from 50 coastal lagoons across the globe ranging from 0 to 24,000,000 items  $\text{m}^{-3}$ . The MPs contamination have been detected in plantae (Chlorophyta), invertebrates (mollusks, annelids, arthropods), and vertebrates (fish, amphibians, birds) [19,57,80]. Once MPs enter the food chain via its accumulation in aquatic organisms, the trophic transfer can cause its ingestion by higher trophic level organisms including humans through food [104,17,78,79,81,96].

Microplastics tend to bioaccumulate in the fish body, leading to deleterious effects [13,37]. In terms of MP composition, polyethylene (PE), polyester, polyamide (PA), polypropylene (PP), polyethylene terephthalate (PET), polyvinyl chloride (PVC), polystyrene (PS), cellophane, rayon, polymethyl methacrylate (PMMA), polytetrafluoroethylene (PTFE) and cellulose and in terms of MP shape, fibres, fragments, films and pellets have been detected in fishes bodies [6]. Smaller MPs have a greater chance of bioaccumulation in fish tissues. This is due to the size-dependent translocation potential of smaller MPs across living cells to the circulatory system. As translocation increases, the extent of bioaccumulation to secondary organs occurs, thereby extending their impact into a wide range of organ systems ([41,47]; des [25,104]). Recently, an accumulation of very small environmental MPs ( $< 3 \mu\text{m}$ ) has been detected in the intestine and muscle of adult benthopelagic fish *Serranus scriba*, caught along Tunisian coasts, which highlights the possible human health risks [112]. Moreover, laboratory studies on MPs exposure in fish confirmed the accumulation of small MPs of size ranging from 1 to 10  $\mu\text{m}$  in the intestine, gills, liver and brain [1,22,26,44,99]. Previous studies reported that MPs induce oxidative damage, behavioural changes, changes in haematological parameters, neurotoxicity, reproductive disruption, altered gene expression, immune impairment, abnormal metabolism and gut microbiota dysbiosis in fish [13,39,46,75]. Smaller MPs (1–10  $\mu\text{m}$ ) were also found to cause significant alterations in antioxidant enzyme activities and oxidative damage [1,26,76,99], lysozyme activity [23], inhibition of acetylcholinesterase (AChE) activity [26,71,8,84], changes in cytochrome P450 (CYP) enzyme activities [44] and reproductive impairment [101,76,99] in fish. Smaller plastic particles of size  $< 10 \mu\text{m}$  can be more toxic than larger plastic particles [37]. Though there are several studies on the effects of virgin MPs in fish, a better understanding is required regarding the consequences triggered by environmentally relevant concentrations of smaller MPs in fish. This is because smaller particles are of most concern for penetration into deeper tissues [55,56]. Furthermore, this study pioneers new insights into the effects of environmentally relevant concentrations of smaller MPs in the brain GNRH activity, plasma vitellogenin level, plasma cortisol level, gene expression of *CYP1A* and *P53* as well as innate immune parameters of fish such as serum lysozyme activity, serum TNF- $\alpha$  level, myeloperoxidase (MPO) activity, respiratory burst activity and serum IgM levels of *Oreochromis niloticus*.

Several field investigations reported environmental concentrations of MPs in aquatic systems as  $\leq 1 \text{ mg/L}$  [100,11,34,59,72,90–92]. So we selected  $\leq 1 \text{ mg/L}$  of MPs as the exposure concentrations along the 14 days exposure period. Nile tilapia (*Oreochromis niloticus*), a suitable candidate for aquaculture in freshwater and brackish water, is the third most cultured finfish [29]. Since MP pollution is very prevalent in inland aquaculture [111,15,89], the large scale culture of *Oreochromis niloticus* may expose the fish to MP pollutants in the water. Hence, we have

selected *O. niloticus* as the candidate species to study the effect of environmentally relevant concentrations of microplastics [11,21,28,73,97]. In particular, this study is aimed to assess the accumulation of environmentally relevant concentrations of smaller MPs in different tissues and its effects on the physiology and immunity of tilapia fish. Polystyrene microplastics (PS MPs) of size 1  $\mu\text{m}$  were used in this study. The MP accumulation in the gills, intestine, muscle, liver, spleen, gonads and brain was investigated. Hematological parameters, serum biochemical parameters, oxidative stress parameters, immunological parameters, cortisol level, AChE activity, gonadotropin-releasing hormone (GNRH) activity, vitellogenin level and gene expression were analysed to evaluate the toxicity of MPs.

## 2. Materials and methods

### 2.1. MP-Characterization

PS MPs of 1  $\mu\text{m}$  size were purchased from Sigma-Aldrich, supplied as an aqueous suspension with a concentration of 10% and a density of 1.05  $\text{g/cm}^3$ . The PS-MPs suspension was prepared with deionized water and sonicated before every usage. The size and the composition of the microplastic were confirmed by the scanning electron microscope (SEM) and Fourier- Transform infrared (FTIR) spectrophotometry respectively. SEM analysis was done using Field Emission-Scanning Electron Microscope (FE-SEM, Joel JSM 5400) at STIC-Sophisticated Test & Instrumentation Centre, CUSAT, Kerala) and FTIR spectra were recorded using a Perkin Elmer Spectrum 100 spectrometer (Inter University Instrumentation Centre, MG university, Kerala).

### 2.2. Experimental design

One hundred and fifty freshwater fish *Oreochromis niloticus* were procured from a private fish farm. The fish were acclimatized for 2 weeks to the laboratory conditions, during which they were fed twice daily with artificial feed. After acclimatization, fish were (with an initial body weight of  $(27.59 \pm 1.34 \text{ g})$  (mean  $\pm$  SD)) randomly distributed into 4 groups in four replicates. Seven fish were kept per tank containing 15 L of water (each tank with a capacity of 20 L). The first group was kept as control (denoted as C). Fishes of the 2nd, 3rd and 4th groups were exposed to PS MP at the concentrations 0.01 (denoted as A), 0.1 (denoted as B) and 1  $\text{mg/L}$  (denoted as D) respectively. Continuous aeration was provided to maintain the complete dispersion of the MP particles in the water. The semi static MP-exposure experiment lasted for 14 days, during which the exposure medium was renewed every 3 days to maintain MP concentration and for the removal of waste.

### 2.3. Sample collection

At the end of the 14 days of exposure, randomly selected fishes from four groups with each of two samples constituting to a sample size of  $N = 8$  for one treatment were sampled and euthanised by MS 222 (Sigma-Aldrich, USA). Blood samples were collected from the caudal vein, using a 1 ml syringe and transferred into a microcentrifuge tube containing anticoagulant (heparin) for immunological analysis. For serum biochemical and enzymatic analysis, blood was collected in anticoagulant-free syringes and transferred to microcentrifuge tubes and allowed to clot at room temperature for 30 min and then in chilled condition for 4 hrs. Serum was separated from coagulated blood by centrifugation at 3000 rpm for 15 min at  $4^{\circ}\text{C}$  and thereafter stored at  $-20^{\circ}\text{C}$  for further analysis. Plasma was collected using ethylenediaminetetraacetic acid (EDTA) (Sigma-Aldrich, USA) as an

anticoagulant after centrifugation at 3000 rpm for 15 min at 4° C.

Intestine, liver, spleen and muscle tissue samples were collected and kept at – 20 °C under sealed conditions to analyse the bioaccumulation of MP. Liver and brain tissues were immediately dissected and weighed to estimate tissue enzymatic activity. Each tissue sample was homogenized in phosphate-buffered saline (PBS) (1:9) with a mortar and pestle on ice. The homogenates were then centrifuged for 10 min at 10000 rpm to get the supernatant. The clear tissue suspensions were stored in microcentrifuge tubes at – 80° C for enzyme analysis. Gills, intestine, liver and spleen tissues were fixed in 10% neutral buffered formalin (NBF) for histological analysis. Liver (for the estimation of *CYP1A*, *HSP70* and *P53* gene expression) and spleen (for the estimation of *HSP70* gene expression) were aseptically collected from the euthanized fish and stored in RNA later (Sigma-Aldrich, USA) (at –20° C for later analysis).

## 2.4. Bioaccumulation

The intestine, gills, liver, spleen, muscle, gonad and brain were selected to analyse the accumulation of MPs. Based on the method described by Deng et al. (2017) approximately 100 mg of each tissue sample was digested in 1 ml of H<sub>2</sub>O<sub>2</sub> (HiMedia, India) (30% V: V) at 70°C for 2 hrs, then diluted with deionized water to a final volume of 5 ml from which 100 µl resultant solution was then examined after fixing in a clean glass slide and counted under the light microscope.

## 2.5. Histopathological analysis

The NBF fixed gills, intestine, liver and spleen tissues of fish were embedded in paraffin wax, sectioned at 4 µm thickness, and stained with hematoxylin and eosin (H&E) (HiMedia, India) for final observation.

## 2.6. Blood analysis: hematological, biochemical and immunological parameters

The whole blood was analysed for hematological parameters such as red blood cells (RBCs), white blood cells (WBCs), haemoglobin (Hb), haematocrit (HCT) and platelet count (PLT), using Mindray Auto-Veterinary Hematology analyser (BC2800, China).

Serum biochemical parameters like glucose, total protein, albumin/globulin (A/G) ratio, serum glutamic-oxaloacetic transaminase (SGOT), serum glutamic-pyruvic transaminase (SGPT) and alkaline phosphatase (ALP) were estimated using Agappe Diagnostic kits (Agappe Diagnostics Ltd, Kerala, India). The parameters were determined according to the manufacturer's instructions and the absorbance was read at related wavelengths using spectrophotometer.

Innate immune parameters such as respiratory burst activity in heparinized blood and lysozyme activity, MPO activity and TNF-α in serum and adaptive immune parameter IgM in serum were analysed.

Respiratory burst activity was performed following the nitro blue tetrazolium (NBT) assay protocol of Anderson and Siwicki [4]. 100 µl of heparinized blood and an equal volume of 0.2% NBT (Sigma-Aldrich, USA) in phosphate-buffered saline solution) were placed into a micro-titer plate well and co-incubated for 45 min at room temperature. Fifty microliters from the NBT-blood cell suspension were added to 1 ml of N, N-dimethylformamide (HiMedia, India) and centrifuged at 2000 rpm for 10 min. The optical density (OD) of the supernatant was measured using UV-Visible spectrophotometer (HACH, Germany) at 540 nm against the blank in which the blood was replaced with distilled water.

The serum lysozyme activity was assayed, according to the turbidimetric assay described by Demers and Bayne [24], which measures the lysis of a suspension of *Micrococcus lysodeikticus* (Sigma-Aldrich, USA) (0.75 mg/ml) in 0.1 M sodium phosphate buffer (pH 6.2). 25 µl of each serum sample and dilutions of the hen egg white lysozyme standard (Sigma-Aldrich, USA) 0–20 µg/ml of 0.1 M sodium phosphate buffer with pH 6.2) were placed into wells of a flat bottomed, 96-well plate. Then, 175 µl of the bacterial suspension was added to each well. After

rapid mixing, the rate of reduction in absorbance was read against *M. lysodeikticus* blank every 30 s for 5 min at 450 nm on a microplate reader (Bio-Rad, Japan). The lysozyme activity was expressed in µg/ml based on a standard curve.

Total serum myeloperoxidase activity was determined following the protocol of Quade and Roth [77] and minor modifications incorporated by Sahoo et al. [83]. 10 µl serum was diluted with 90 µl of Hank's balanced salt solution (HBSS) without Ca<sup>2+</sup> or Mg<sup>2+</sup> (Sigma-Aldrich, USA) in a 96-well microtitre plate. 35 µl of 20 mM 3,3',5,5'-tetramethyl benzidine hydrochloride (TMB) (Genei, India) and 5 mM H<sub>2</sub>O<sub>2</sub> were added and incubated for 2 min. The reaction was stopped by adding 35 µl of 4 M sulphuric acid (HiMedia, India). The OD was determined in a microplate reader at 450 nm.

The serum TNF-α and IgM levels were measured by commercial fish ELISA kits supplied by Origin Diagnostics and Research, India following the protocols of the manufacturer.

## 2.7. Estimation of oxidative stress-related biomarkers

Oxidative stress parameters like superoxide dismutase (SOD), catalase (CAT) and malondialdehyde (MDA) were measured in liver tissue homogenate.

SOD was assayed according to the method described by [68] based on the oxidation of epinephrine adrenochrome transition by the enzyme. Fifty microliters of the sample were taken in the cuvette and 1.5 ml 0.1 M carbonate- bicarbonate buffer containing 57 mg/dl EDTA (pH 10.2) and 0.5 ml 3 mM epinephrine (HiMedia, India) was added and mixed well. Change in optical density at 480 nm was read immediately for 3 min in a UV spectrophotometer. One unit of SOD activity was the amount of protein required to give 50% inhibition of epinephrine auto oxidation. SOD is expressed as a unit activity (amount of protein required to give 50% inhibition of epinephrine auto oxidation).

CAT was assayed according to the method described by Takahara et al. [93]. To a reaction mixture of 2.45 ml phosphate buffer (50 mM, pH 7.0), an enzyme source was added and the reaction was started by the addition of 1.0 ml of H<sub>2</sub>O<sub>2</sub> solution. The decrease in absorbance was measured at 240 nm at 15 s intervals for 3 min. The enzyme blank was run simultaneously with 1.0 ml distilled water instead of H<sub>2</sub>O<sub>2</sub> solution. Enzyme activity was expressed as nano molecules H<sub>2</sub>O<sub>2</sub> decomposed/min/mg protein at 37 °C.

The protein content in liver tissue was determined following the protocol of Lowry et al. [65].

The MDA levels were determined using the ELISA kit purchased from Origin Diagnostics and Research, India based on the manufacturer's instructions.

## 2.8. Estimation of cortisol level, AchE activity, GNRH and vitellogenin

The plasma cortisol level, brain AchE activity, brain GNRH activity and plasma vitellogenin level were detected by using commercial ELISA kits purchased from Origin Diagnostics and Research, India according to the manufacturer's protocol.

## 2.9. Analysis of gene expression

Total RNA was extracted from the liver and spleen with TRI Reagent (Sigma-Aldrich, USA) and quantified with a Qubit 4 Fluorometer (Invitrogen, USA). RNA was then treated with DNase I (amplification grade, 1 unit/µg RNA) (Invitrogen, USA), and cDNA synthesis kit with RNase inhibitor (Origin Diagnostics and Research, India) was used to synthesize cDNA from 2 µg of total RNA, following the manufacturer's instructions. Quantitative real-time PCR was performed in CFX96 Real-time PCR Detection System (Bio-Rad, USA) with SYBR green PCR master mix (Origin Diagnostics and Research, India). The sequences of primers for the target genes and internal standard gene (*EF-1α*) are shown in Table 1. The thermocycling conditions for the reaction were 30 s at 95°

**Table 1**

The sequences of primers for the target genes and internal standard gene (*EF-1 $\alpha$* ).

| Gene name                      | Primer sequence (5'→3')                               | References       |
|--------------------------------|-------------------------------------------------------|------------------|
| <i>HSP70</i>                   | F: CATCGCCTACGGTCTGGACAA<br>R: TGCCGTCTTCAATGGTCAGGAT | [60]             |
| <i>CYP1A</i>                   | F: CGTCGTCTCTCTGTTGCC<br>R: CATCGTCGTGGTGGTCATAGC     | [60]             |
| <i>P53</i>                     | F: GCATGTGGCTGATGTTGTTTC<br>R: GCAGGATGGTGGTCATCTCT   | [30]             |
| <i>EF-1<math>\alpha</math></i> | F: CTGTACCGTCGGTCGTGTT<br>R: GGTGCATCTCCACGGACTTC     | Bhat et al. [12] |

C, followed by 40 cycles of 5 s at 95<sup>0</sup> C and 1 min at 60<sup>0</sup> C. The relative fold changes in gene expression were analysed according to the 2<sup>- $\Delta\Delta$ Ct</sup> method [63].

### 2.10. Statistical analysis

The statistical evaluation was carried out by Statistical Program for Social Sciences (SPSS version 22, SPSS Inc., IL, USA). Analysis were carried out in quadruplicate and the means of all parameters were measured underlying all the assumptions of ANOVA for significance ( $p \leq 0.05$ ) by ANOVA (analysis of variance). Duncan's multiple-range test was performed to determine differences among treatments with significance set at  $P < 0.05$  or  $P < 0.01$ . All values are expressed in terms of mean  $\pm$  standard error of the mean (SEM). Different letters represent significant differences between treatment means.

## 3. Results

### 3.1. Characterization of PS MP

The SEM image of PS MP (Fig. 1) used in this study confirmed that the MPs are spherical in shape with an average diameter of 995.412 nm (0.995  $\mu$ m). FTIR spectroscopy revealed that the MP is composed of polystyrene (Fig. 2).

### 3.2. Accumulation of PS MPs in different tissues

There was no mortality or abnormality observed during the PS MP exposure. Accumulation of PS MPs was observed in the intestine, gills, liver, spleen, muscle, gonad and brain of tilapia exposed to all three MP concentrations (0.01, 0.1 and 1 mg/L). The PS-MPs accumulation in tilapia tissues was found to follow the order of intestine > gills > liver > spleen > muscle  $\approx$  gonad > brain (Figs. 3 and 4).

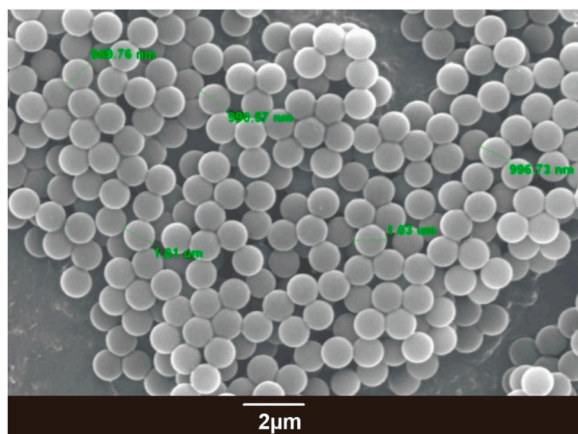


Fig. 1. A scanning electron microscope (SEM) image (x7000) of PS-MP.

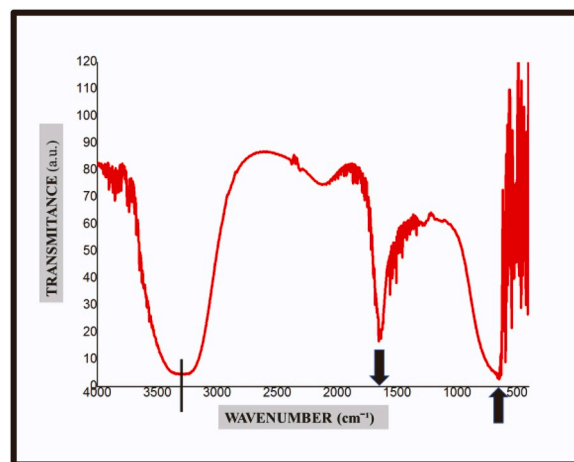


Fig. 2. FTIR spectroscopy for PS-MP.

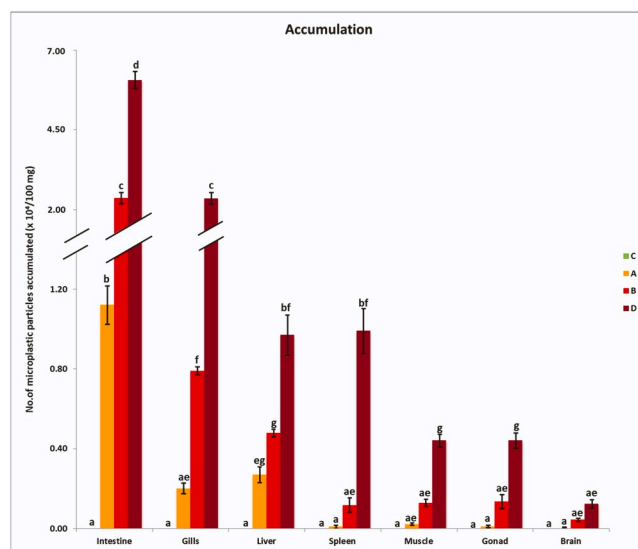


Fig. 3. Accumulation of PS-MP Particles ( $\times 10^4/100$  mg) in *O. niloticus* tissues after 14 days exposure ( $p < 0.05$ ). A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.

### 3.3. Histopathology

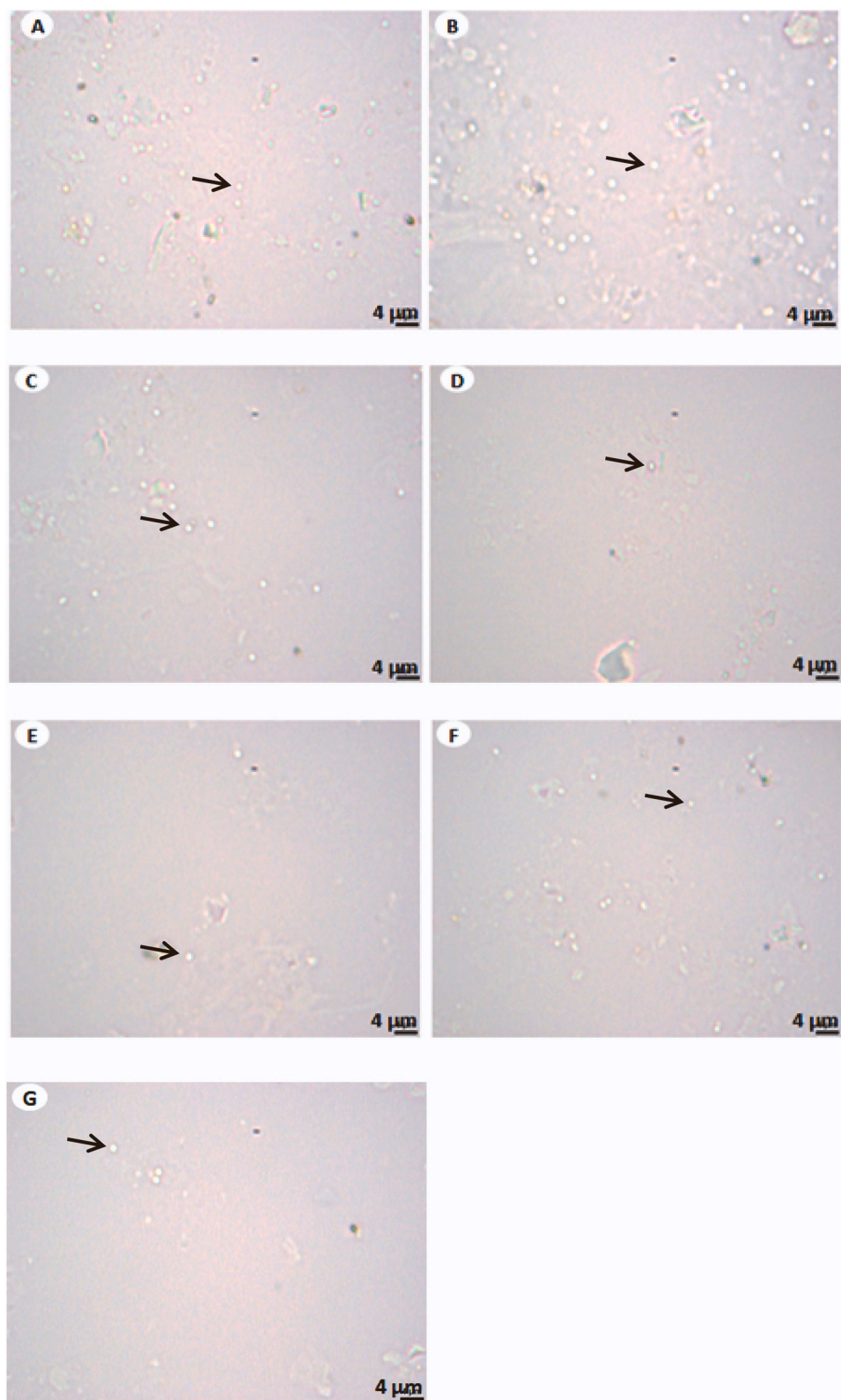
The 14 days PS-MPs exposure did not cause any changes in the normal tissue morphology of the gills, intestine, liver and spleen of Nile tilapia.

### 3.4. Hematological parameters

The PS MP exposure was found to cause variations in the hematological parameters of *O. niloticus*. A significant decrease in RBC, Hb and HCT values was observed in tilapia exposed to the highest two concentrations of MP (0.1 and 1 mg/L). Whereas, WBC exhibited a significant increase in 0.1 and 1 mg/L MP exposed groups and PLT showed a significant increase in the 1 mg/L MP exposed group as compared to the control (Table 2).

### 3.5. Serum biochemical parameters

The serum biochemical parameters such as glucose ( $p < 0.01$ ), total protein, A/G ratio, SGOT ( $p < 0.05$ ) and ALP ( $p < 0.01$ ) showed a



**Fig. 4.** Microscopic images (x100) of MPs in fish tissues after 14 days of exposure. A-Gills, B-Intestine, C-Liver, D-Muscle, E-Brain, F-Spleen and G-Gonad.

significant increase after exposure to 0.01, 0.1 and 1 mg/L of PS MPs for 14 days compared to the control group in dose dependent manner. But serum SGPT level showed a significant ( $P < 0.01$ ) increase only in the 1 mg/L MP exposed group in comparison with the control group (Fig. 5).

### 3.6. Oxidative stress parameters

SOD activity in the liver of MP exposed tilapia reduced significantly in comparison with the control ( $P < 0.05$ ). However, the reduction was not significant among the MP treatments. Even though the MP exposures slightly increased hepatic CAT activity in a dose dependent manner, no

**Table 2**

Effect of 14 days MPs exposure on the hematological parameters of *O. niloticus* ( $p < 0.05$ ).

| Parameters                          | Control              | 0.01 mg/L            | 0.1 mg/L             | 1 mg/L               |
|-------------------------------------|----------------------|----------------------|----------------------|----------------------|
| WBC ( $\times 10^3 / \mu\text{l}$ ) | 116.5<br>$\pm 0.38a$ | 117.3<br>$\pm 0.38a$ | 122.0<br>$\pm 0.95b$ | 127.4<br>$\pm 0.35c$ |
| RBC ( $\times 10^6 / \mu\text{l}$ ) | 1.30 $\pm 0.03a$     | 1.22<br>$\pm 0.02ab$ | 1.13 $\pm 0.02b$     | 0.97 $\pm 0.04c$     |
| Hb (g/dL)                           | 9.1 $\pm 0.02a$      | 9.1 $\pm 0.03a$      | 8.9 $\pm 0.03b$      | 8.7 $\pm 0.02c$      |
| HCT (%)                             | 17.1 $\pm 0.06a$     | 17.1 $\pm 0.09a$     | 16.7 $\pm 0.09b$     | 16.1 $\pm 0.09c$     |
| PLT ( $\times 10^3 / \mu\text{l}$ ) | 285.3<br>$\pm 0.33a$ | 285.3<br>$\pm 0.88a$ | 287.3<br>$\pm 0.33a$ | 290.3<br>$\pm 0.88b$ |

significant differences were observed among the treatments ( $p > 0.05$ ). The MP exposures significantly elevated the liver MDA contents after 14 days of exposure ( $p < 0.01$ ) and the level of elevation was according to the MP exposure concentrations (Fig. 6).

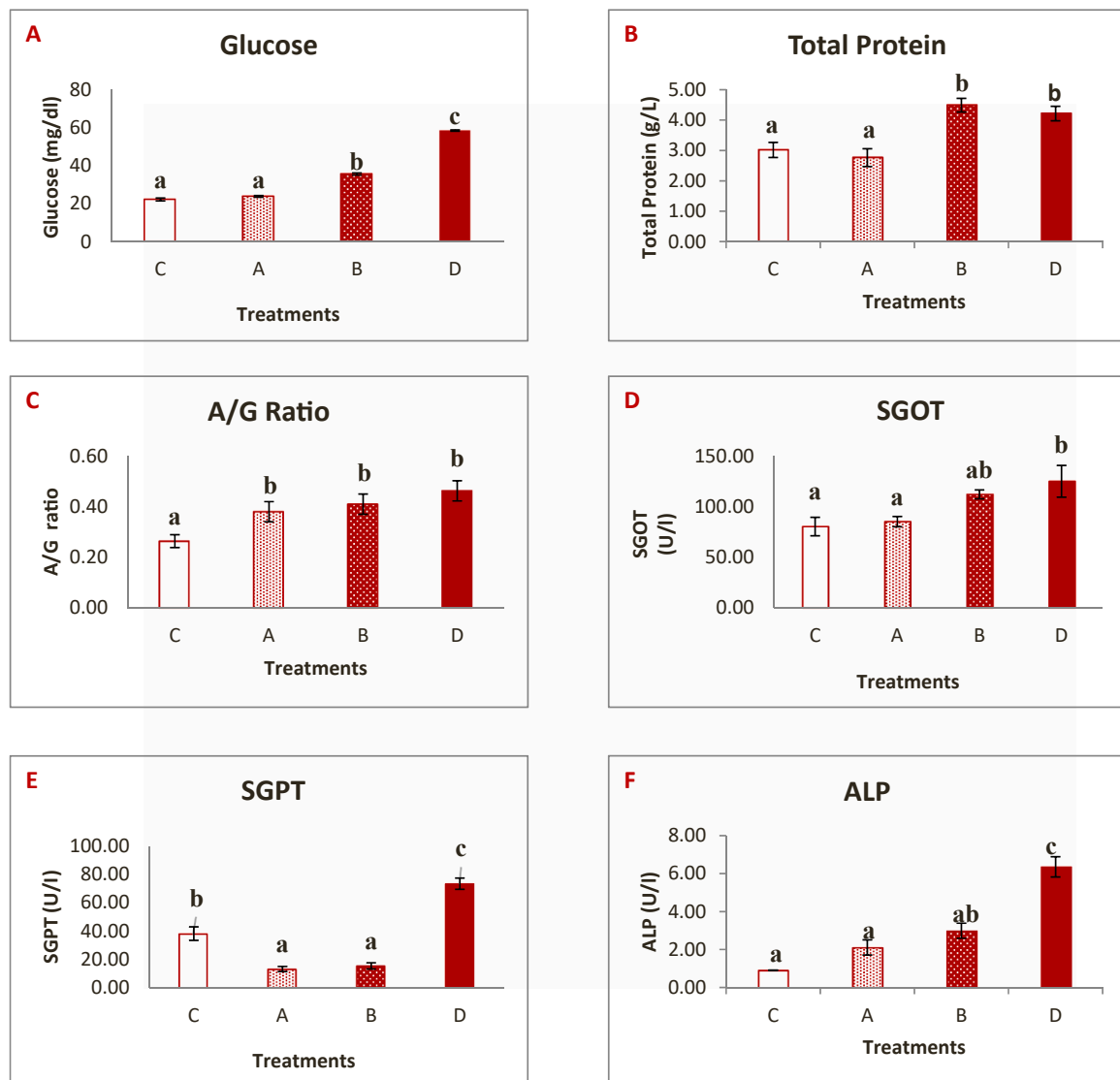
### 3.7. Immunological parameters

The respiratory burst activity significantly increased after exposure

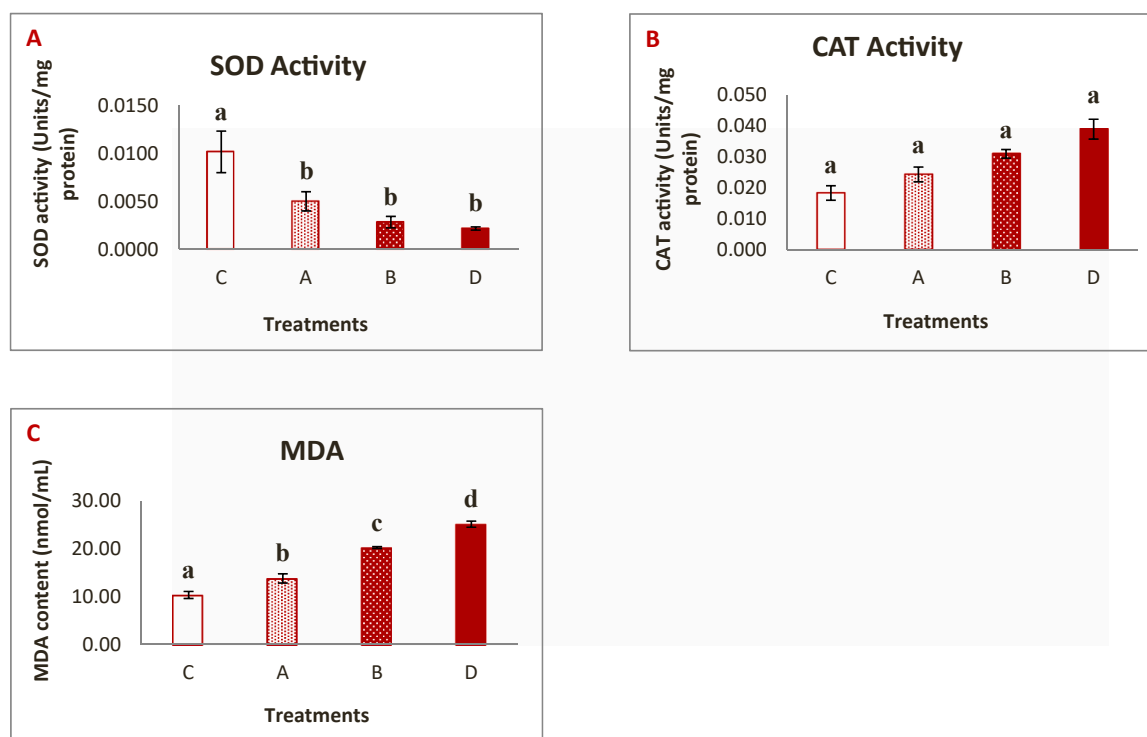
to the 0.1 and 1 mg/L MPs ( $p < 0.05$ ). The serum lysozyme activity exhibited no significant difference among treatments, but there is an increasing trend in the level of activity ( $P > 0.05$ ). Although there was no significant difference in MPO activity levels between the control and 0.01 mg/L MP treatment, the activity was significantly higher in the 0.1 and 1 mg/L MPs exposed group ( $p < 0.05$ ). The PS MPs significantly increased the serum IgM and TNF- $\alpha$  levels in all three treatment groups after the exposure period ( $p < 0.01$ ) in a concentration dependent manner (Fig. 7).

### 3.8. Effect on cortisol level, AChE activity, GNRH and vitellogenin

The plasma cortisol level exhibited a significant increase in response to MP exposure ( $p < 0.01$ ). In contrast, MP exposure significantly inhibited brain AChE activity ( $p < 0.01$ ), brain GNRH ( $p < 0.05$ ) activity and plasma vitellogenin level ( $p < 0.01$ ) after 14 days of exposure in a dose dependent manner. Although a decreasing trend in vitellogenin levels was noticed, the differences were not significant among MP exposed groups (Fig. 8).



**Fig. 5.** Serum biochemical parameters such as glucose ( $p < 0.01$ ), total protein ( $p < 0.05$ ), A/G ratio ( $p < 0.05$ ), SGOT ( $p < 0.05$ ), SGPT ( $p < 0.01$ ) and ALP ( $p < 0.01$ ) of *O. niloticus* exposed to 0, 0.01, 0.1 and 1 mg/L of PS-MPs concentration over a period of 14 days: A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.



**Fig. 6.** Oxidative stress parameters such as SOD ( $p < 0.05$ ), CAT ( $p > 0.05$ ) and MDA ( $p < 0.01$ ) of *O. niloticus* exposed to 0, 0.01, 0.1 and 1 mg/l of PS-MPs concentration over a period of 14 days: A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.

### 3.9. Gene expression

Exposure to PS MP significantly down-regulated ( $p < 0.01$ ) the mRNA expression of *CYP1A* in the livers of all the MP exposed groups compared to the control. However, down-regulation was not significant between the MP treatments. An increase in the PS MP concentration significantly up-regulated the expression of *HSP70* in the liver ( $P < 0.05$ ). Similarly, *HSP70* expression was up-regulated in the spleen tissues. However, only the highest dose (1 mg/L) exposure of PS-MPs had a significant effect on the *HSP70* expression in the spleen in comparison with the control ( $P < 0.05$ ). The expression of *P53* in the liver was found to be downregulated ( $P < 0.05$ ) in 0.01 mg/L MPs concentration whereas, the higher PS-MPs exposure concentrations of 0.1 and 1 mg/L significantly ( $P < 0.05$ ) upregulated the expression of *P53* compared to control samples (Fig. 9).

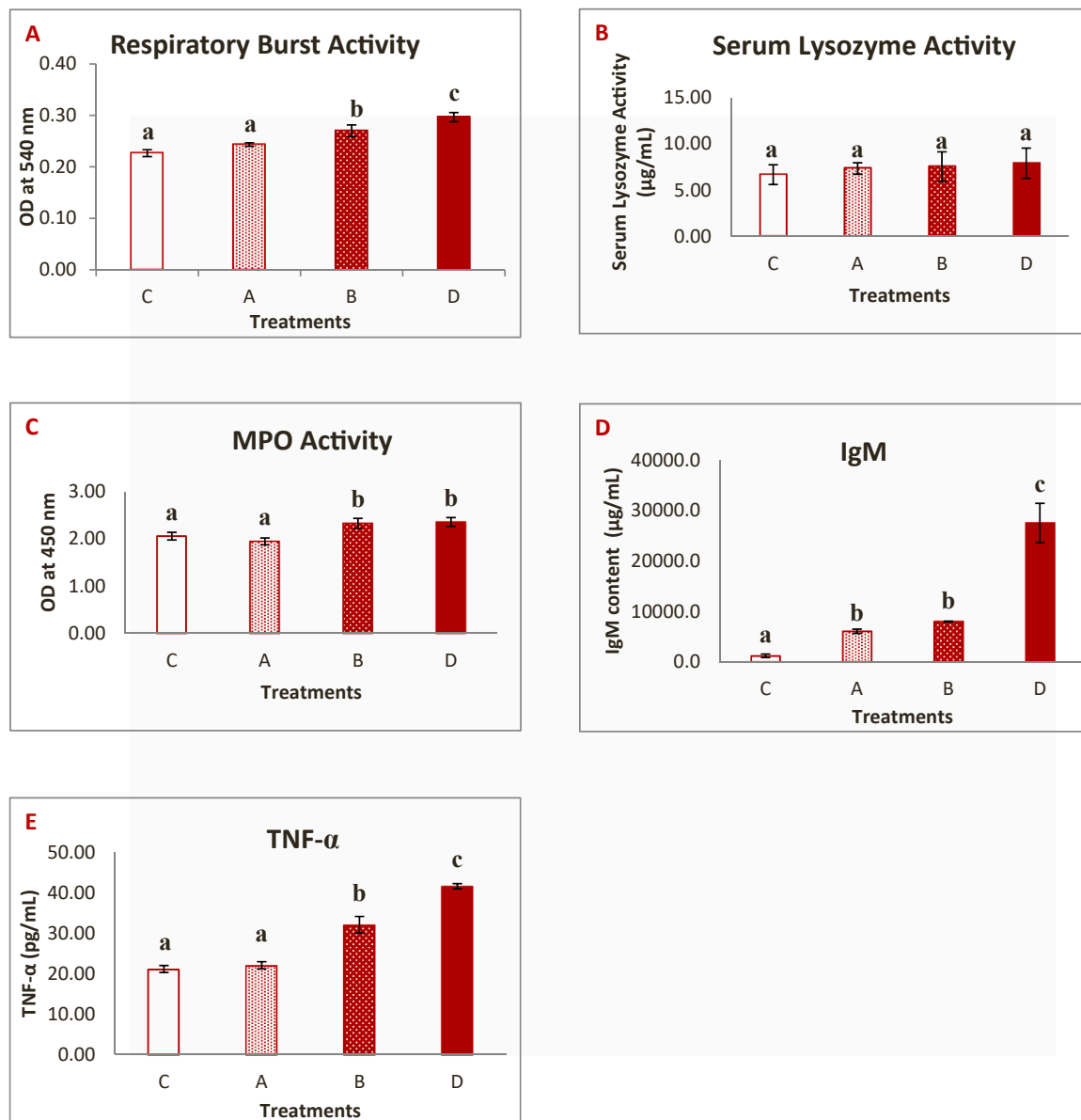
## 4. Discussion

The present study confirmed a dose-dependent MP accumulation in the intestine, gills, liver, spleen, muscle, gonad and brain of Nile tilapia. The polystyrene MPs of sizes 0.25  $\mu\text{m}$  and 8  $\mu\text{m}$  were accumulated in the gills, intestine and liver of goldfish. The microplastic concentration in tissues increased with exposure time [1]. In adult zebrafish, the highest quantity of MPs accumulated in the gastrointestinal tract (GIT) followed by the gill after 7 days of polystyrene MP (5  $\mu\text{m}$ ) exposure at 0.001–20 mg/L. Moreover, no microplastics were observed in the brain and muscles [22]. The accumulation of 5  $\mu\text{m}$  polystyrene MP in red tilapia was highest in the gut followed by the liver, gills and brain after 14 days of exposure at 100  $\mu\text{g/L}$  [26]. Meanwhile, the polystyrene MPs with size 32–40  $\mu\text{m}$  showed a higher accumulation in the gills of the guppy than in the gut [42]. In red tilapia, the accumulations of polystyrene MPs with size 5  $\mu\text{m}$  followed the order of liver > brain > gills > gut after 14 days of treatments with 10  $\mu\text{g/L}$  exposure concentration [44]. In addition, small environmental MPs of size 0.45–3  $\mu\text{m}$  were detected in the intestine and muscle of adult benthopelagic fish *Serranus*

*scriba*, caught along Tunisian coasts [112]. The dietary exposure of early-juvenile European sea bass *Dicentrarchus labrax* with environmental MPs (0.45 –  $\geq 3 \mu\text{m}$ , 3 and 5 days) resulted in MP accumulation in gills, gut and liver. The accumulation of smaller-sized environmental MPs (0.45 – 1.2  $\mu\text{m}$ ) was highest in the liver [113]. However, in this study, the MP (1  $\mu\text{m}$ ) accumulation was highest in the intestine. It is important to note that tissue translocation of small particles is more efficient than larger particles and particles < 20  $\mu\text{m}$  may penetrate cell membranes (Hale et al., 2020- A global perspective on microplastics). The 1  $\mu\text{m}$  PS MPs used in our study were small enough to be transported and accumulated in different organs.

No obvious histological changes were observed in the gastrointestinal tract (GIT) of zebrafish after 7 days exposure to PS MPs of size 5  $\mu\text{m}$  at concentrations  $\leq 1 \text{ mg/L}$  [22]. Similarly, a previous study reported that 14 days of exposure to PS MPs (0.5  $\mu\text{m}$ ) at 100 and 1000  $\mu\text{g/L}$  did not alter the histology of the guts of zebrafish [49]. Short-term exposure to low-density polyethylene (LDPE) fragments (<17.6  $\mu\text{m}$ ) for 20 days at environmentally-relevant concentrations (500  $\mu\text{g/L}$ ) had no impacts on the histopathology of the gill, liver, brain, kidney, and intestine in the larvae of zebrafish [51]. Similarly, in the present study, the environmentally-relevant concentrations of MPs exposure for a short-term duration displayed no significant changes in the histology of gills, intestine, liver and spleen of Nile tilapia. Interestingly, Gu et al. [36] also proved the inability of environmentally-relevant concentrations of smaller MPs to induce histological alterations in the fish intestines. However, Lu et al. [66] and Zhang et al. [109] reported contrary results. In our opinion, short-term exposure to environmentally-relevant low concentrations of smaller MPs would not alter histology.

No significant effects on haematological parameters such as RBC, WBC, Hb and HCT were reported in juvenile European seabass after 10 days of exposure to microplastics (0.25 mg/L) [23]. There were no changes in WBC, PLT and HCT values of Nile tilapia after 15 days of MPs exposure at a concentration of 1 mg/L. However, a significant decrease in RBC and Hb levels was noted [39]. Similarly, in the present study,



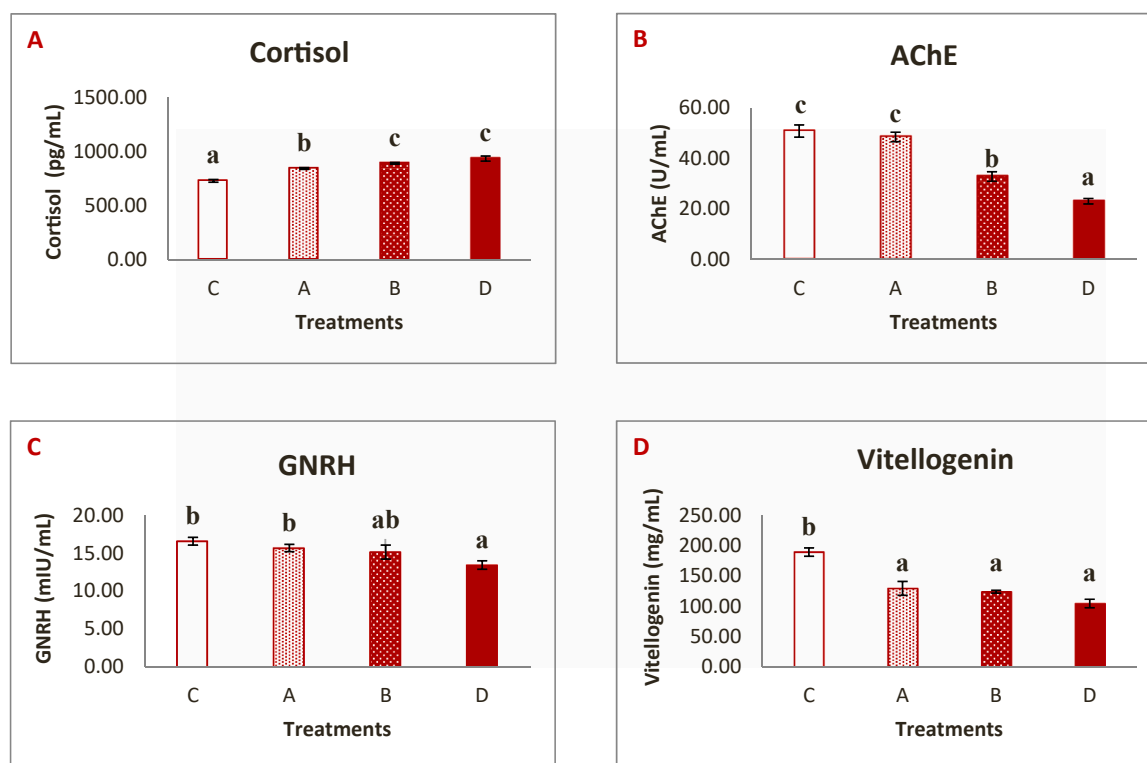
**Fig. 7.** Immunological parameters such as Respiratory Burst Activity ( $p < 0.05$ ), Serum Lysozyme Activity ( $p > 0.05$ ), MPO Activity ( $p < 0.05$ ), IgM ( $p < 0.01$ ) and TNF- $\alpha$  ( $p < 0.01$ ) of *O. niloticus* exposed to 0, 0.01, 0.1 and 1 mg/L of PS-MPs concentration over a period of 14 days: A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.

exposure to MPs reduced RBC, Hb and HCT which might be due to the heam-dilution of blood as a result of MP induced-tissue damage [95] and hemolysis [58]. In addition, the findings of the present investigation depicted an increase in WBC and PLT counts as an enhanced defence mechanism in Nile tilapia against MP exposure. According to previous reports [2,39,86] and our findings, we hypothesized that heam-dilution of blood, hemolysis and anemia are indications MPs intoxication in organisms.

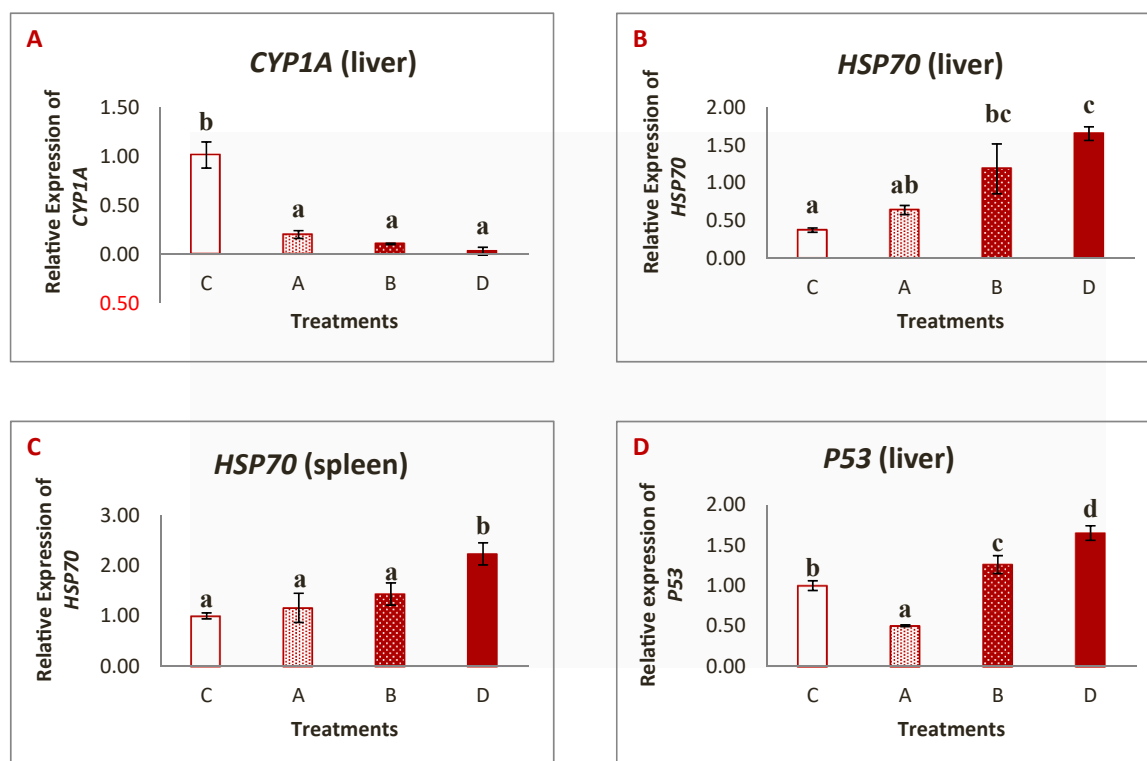
Our results demonstrate that MPs exposure caused alterations in the serum biochemical parameters of Nile tilapia. An increase in blood glucose level is considered to be a secondary marker of stress response [94]. According to our results, MPs at environmentally-relevant concentrations can induce stress in fish. Moreover, in the present study, the increased total protein after MPs exposure might be due to the increase in protein synthesis to meet the MP-induced high energy demand. In addition, the high energy demand during stress accelerates the glycolysis and gluconeogenesis in the fish body to provide glucose and energy [20,98]. An increase in A/G ratio indicates immunosuppression [87]

and was observed in MPs treated fishes. Elevated levels of the hepatic enzymes Aspartate transaminase (AST), Alanine transaminase (ALT) and ALP indicate hepatic damage [10,14,64]. Our observations showed that MPs accumulation in liver induced hepatic damage. Glucose, AST, ALT and ALP levels were increased significantly in the plasma of common carp juveniles exposed to MPs. However, the levels of total protein, albumin and globulin were significantly decreased in MP treated groups [7]. The 21 days MPs exposure on juvenile common carp caused a decrease in total protein, albumin and globulin levels and no alterations in glucose, AST, ALT and ALP levels in plasma [70]. Wen et al. [102] reported an increase in ALP levels in discus fish after 30 days of exposure to polystyrene MPs (32–40 µm; 50 and 500 µg/L). MPs exposure at a concentration of 1 mg/L for 15 days resulted in a significant elevation of ALT, ALP, glucose and A/G ratio and no changes in AST and total protein in early juvenile Nile tilapia [39].

In this study, polystyrene MP caused an imbalance in the antioxidant defence of tilapia. No significant alterations were found in the hepatic SOD and CAT activities of adult zebrafish after 1 mg/L of polystyrene



**Fig. 8.** Effect on cortisol level ( $p < 0.01$ ), AChE activity ( $p < 0.01$ ), GNRH ( $p < 0.05$ ) and vitellogenin ( $p < 0.01$ ) of *O. niloticus* exposed to 0, 0.01, 0.1 and 1 mg/l of PS-MPs concentration over a period of 14 days: A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.



**Fig. 9.** Effects on the expressions of CYP1A in liver ( $p < 0.01$ ), HSP70 in liver and spleen ( $p < 0.05$ ), and P53 in liver ( $p < 0.05$ ) of *O. niloticus* exposed to 0, 0.01, 0.1 and 1 mg/l of PS-MPs concentration over a period of 14 days: A, control without PS MP exposure; B, 0.01 mg/L PS MP exposure; C, 0.1 mg/L PS MP exposure; D, 1 mg/L PS MP exposure.

MP (5  $\mu\text{m}$ ) exposure for 7 days [22]. Chronic dietary exposure (one-month and two-month) of polyvinyl chloride (PVC) MPs in larvae of common carp *Cyprinus carpio* var. significantly decreased SOD activity in the liver, intestine and gills [105]. In contrast, the polystyrene MP exposure was found to induce the antioxidant defense of fish. In goldfish, exposure to polystyrene for 28 days induced SOD and CAT activity [1]. Similarly, a significant increment in hepatic SOD activity was obtained in the 5  $\mu\text{m}$  polystyrene MP exposed red tilapia at 100  $\mu\text{g/L}$  for 14 days [26]. The SOD and CAT activities increased significantly in a concentration-dependent manner in the viscera of juvenile guppy after 28 days exposed to polystyrene MPs with size 32–40  $\mu\text{m}$  (100 and 1000  $\mu\text{g/L}$ ) [42] and in the liver of early juvenile discus fish after 30 days exposure with the same sized PS MPs (50 and 500  $\mu\text{g/L}$ ) [102]. SOD and CAT activities were decreased in the gut and liver of marine medaka by polystyrene MP (45  $\mu\text{m}$ ) exposure for 14 days [50]. Similarly, the hepatic SOD and CAT activities were diminished in marine medaka after 60 days of PS MPs (10  $\mu\text{m}$ ) exposure to environmentally relevant concentrations [99]. Our study reinforces these findings by reporting the imbalance in antioxidant enzymes in fish exposed to MPs.

MDA, a marker of lipid peroxidation, also increased in the liver tissue after 14 days of MPs exposure, indicating oxidative damage. The 14 days of polystyrene MP exposure with size 5  $\mu\text{m}$  at a concentration of 100  $\mu\text{g/L}$  could significantly elevate the liver MDA contents in red tilapia [26]. Dietary polyethylene MP exposure increased MDA levels in *Sparus aurata* [16]. Likewise, chronic dietary PVC MP exposure (30% of diet, 60 days) significantly enhanced MDA levels in the liver of common carp [105]. Similarly, the polystyrene MPs with size 32–40  $\mu\text{m}$  caused a significant elevation in MDA content in the viscera of juvenile guppy after 28 days of exposure at 100 and 1000  $\mu\text{g/L}$  [42] and in the liver of early juvenile discus fish after 30 days exposure at 50 and 500  $\mu\text{g/L}$  [102]. Hepatic MDA levels were increased in marine medaka exposed to environmentally relevant concentrations of 10  $\mu\text{m}$  sized PS MPs for 60 days [99]. These findings are in agreement with our observation, that MPs exposure at environmentally relevant concentrations induced oxidative stress in fish.

The effects of MPs exposure on innate immune parameters such as respiratory burst activity, lysozyme activity, MPO activity and TNF- $\alpha$  and the adaptive immune parameter IgM were evaluated in this study. Respiratory burst activity is an indicator of nonspecific immunity in fish [82]. MPO is an enzyme present in neutrophils of teleosts which is involved in inflammatory response and microbicidal activity [18,54]. TNF- $\alpha$  is a pro-inflammatory cytokine [38]. IgM plays key roles in host defense, immune regulation and immunological tolerance [62]. In the present study, respiratory burst activity, MPO activity, TNF- $\alpha$  and IgM levels were found to elevate after MPs exposure. Consequently, the fish tried to survive the negative effects of MPs exposure by the stimulation of immunological responses. Fish lysozyme is a bacteriolytic cationic enzyme [3,85]. Our findings showed that the lysozyme activity of Nile tilapia was unaffected by MPs exposure for 14 days. Similarly, there was no change in the lysozyme activity of discus fish after 30 days of PS MP exposure (32–40  $\mu\text{m}$ ; 50 and 500  $\mu\text{g/L}$ ) [102]. The total immunoglobulin and lysozyme activity in juvenile common carp exposed to 250 and 500  $\mu\text{g/L}$  of MPs for 30 days was lower compared to the control group [7]. After 10 days of MPs exposure (0.25 mg/L), the lysozyme activity increased in skin mucus of European seabass juveniles [23]. Increased activity of MPO, a proinflammatory enzyme present in neutrophils has been related to inflammation [52]. In the present study, a notable increase in MPO activity was observed in *O. niloticus* exposed to 0.1 and 1 mg/L MPs treated group. Long-term dietary exposure of polyethylene MP (diet with 10% MP) caused an increase in MPO activity in the gut and liver of gilthead seabream *Sparus aurata* ([88] & [16]). TNF- $\alpha$  level in the gut of polystyrene MP treated guppy showed similar results as this study [43]. The mRNA levels of TNF- $\alpha$  were increased in the livers of the 3 mg/L polystyrene MP treated zebrafish after 12 and 24 days of exposure [106].

AChE plays a vital role in the nervous system and is recognized as an

environmental biomarker [74]. In this study, significant AChE inhibition was observed in 0.1 and 1 mg/L MPs-treated groups. Exposure to 100  $\mu\text{g/L}$  of 5  $\mu\text{m}$  polystyrene MP for 14 days in red tilapia caused significant inhibition of AChE activity to [26]. Limonta et al. [61] also reported AChE inhibition by 1000  $\mu\text{g/L}$  of MPs in zebrafish after 20 days. Significant AChE inhibition was detected in zebrafish larvae exposed to 2 mg/L of MPs (1–5  $\mu\text{m}$ ) for 14 days [84]. Similarly, AChE activity was significantly decreased in discus fish (*Symphysodon aequifasciatus*) and common carp (*Cyprinus carpio*) upon polyethylene MP exposure for 30 days [7,103]. Even 96 h short-term MP exposure at low concentrations caused AChE inhibition in juvenile European seabass (*Dicentrarchus labrax*) [8] and juvenile common goby (*Pomatoschistus microps*) [71]. According to these studies, it is noteworthy that the sensitivity of fish AChE to MP pollution can lead to neurotoxicity.

GnRH regulates the reproduction in fishes through the production and secretion of the gonadotropins (follicle-stimulating hormone, FSH and luteinizing hormone, LH) [69]. Vitellogenin, the yolk precursor protein, is used as a biomarker of fish exposure to estrogenic endocrine disruptors in aquatic environments [40]. Low vitellogenin level indicates reproductive disruption [53]. The current results illustrated that MP exposure reduced the reproductive performance in Nile tilapia through the inhibition of GnRH activity and lowering the vitellogenin level. Ismail et al., [45] reported that MP exposure could adversely affect the reproductive performance of male tilapia by decreasing serum luteinizing hormone (LH) and testosterone (T) levels. Another study [76] demonstrated the reproductive impairment in zebrafish in response to polystyrene MP (1  $\mu\text{m}$  size) exposure for 21 days. The 10  $\mu\text{m}$  polystyrene MP (200  $\mu\text{g/L}$ , 60 days) and 2  $\mu\text{m}$  polystyrene MP (2, 20 and 200  $\mu\text{g/L}$ , 150 days) exposure induced reproductive impairment on marine medaka (*Oryzias melastigma*) by disturbing transcription of genes in the reproductive axis ([99] and [101]).

Results of the present study indicate that 14 days of MPs exposure down-regulated hepatic *CYP1A* gene expression in tilapia. Similar to our observations, two-month chronic dietary exposure to PVC MPs down-regulated *CYP1A* gene expression in the liver of common carp larvae [105]. *CYP1A1* levels in the liver decreased significantly after 12 and 24 days of polystyrene MP exposure in zebrafish [106]. In contrast, significant up-regulation of Cytochrome P450 family gene *CYP2P8* was observed in the head kidney of zebrafish exposed to 100 and 1000  $\mu\text{g/L}$  of MPs for 20 days [61]. The induction of Cytochrome P450 genes suggests that MPs can induce the detoxification metabolism in fish. However, in the present study, higher MP accumulation in the liver induced oxidative damage, which is evident with the corresponding changes in MDA levels. This might have disrupted the hepatic detoxification mechanism resulting in down-regulation of hepatic *CYP1A* gene expression.

In the current study, MP-induced cellular stress is evident from the up-regulation of the *HSP70* gene in the liver and spleen of Nile tilapia. Abarghouei et al. [1] also reported the up-regulation of the *HSP70* gene in goldfish in response to 28 days of polystyrene MP exposure. Our results show that cortisol levels increased in tilapia, following MPs exposure. We found that environmentally-relevant concentrations of MPs exposure to tilapia increased the three tested indicators of stress responses such as the plasma cortisol level, serum glucose level, and liver and spleen *HSP70* mRNA expression. These results are probably due to high cortisol level that could have caused elevated glucose levels and *HSP70* expression during stress conditions in fish [108,9,98]. Overall, MPs exposure was observed to be a stressor in this study. According to previous studies, up-regulated mRNA expression of *HSP70* was found to enhance the immune response under stress [33,110]. We also observed enhanced immunological responses after MPs exposure, probably as a result of up-regulated *HSP70* expression.

Oxidative stress has been found to activate *P53* gene expression which triggers apoptosis [31]. Our results showed a significant up-regulation of the expression of *P53* in the liver of Nile tilapia exposed to 0.1 and 1 mg/L of PS-MPs. This upregulation of *P53* expression reflects

the oxidative stress generated from MP exposure. Significant upregulation of *P53* was observed in the testis of zebrafish exposed to polystyrene MP with size 1  $\mu\text{m}$  (100 and 1000  $\mu\text{g/L}$ ) for 21 days. The upregulation was highest in the 100  $\mu\text{g/L}$  exposure group followed by the 1000  $\mu\text{g/L}$  treatment group [76]. Our results proved that MPs exposure caused oxidative stress mediated activation of the *P53* gene which might initiate apoptosis or cellular death [67].

## 5. Conclusion

This study confirmed the accumulation of PS MPs of size 1  $\mu\text{m}$  in the gills, intestine, liver, muscle, spleen, gonads and brain of *O. niloticus* after 14 days of exposure. This is the first report on the accumulation of environmentally relevant concentrations of smaller MPs in the muscle of *Oreochromis niloticus*. Also, this is the first study regarding the effects of environmentally relevant concentrations of smaller MPs in the innate immune parameters such as MPO activity, respiratory burst activity and serum IgM levels. Fish were able to counteract MP by increasing respiratory burst activity, MPO activity and serum TNF- $\alpha$  and IgM levels. However, the hematological parameters, serum biochemical parameters, oxidative stress parameters, gene expression, increased cortisol and glucose levels, reduced AChE activity, GNRH activity as well as plasma vitellogenin levels represent stress and disruption of the normal physiology of fish. This disruption can contribute to oxidative damage, neurotoxicity, reproductive failure, failure of cellular detoxification mechanism and eventually organ failure. These findings suggest the ecological risk of MP pollution to the sustainability of the fish population in freshwater ecosystems as well as the potential threat to human health posed by the consumption of fish muscle accumulated with smaller MPs. Moreover, we recommend further studies on the MP size and shape dependent tissue accumulation and translocation to target organs which would help to elucidate the severity of impacts at different tissue and organ levels.

## Environmental implications

Present study analyses the accumulation and toxic effects of low concentrations (0.01–1 mg/L) of smaller microplastics (1  $\mu\text{m}$ ) on the freshwater fish tilapia. Our study demonstrated microplastic accumulation in different fish tissues and altered hematological, serum biochemical, immune and stress parameters as well as oxidative damage, neurotoxicity, reproductive failure and failure of cellular detoxification mechanism in response to microplastic exposure. This study highlights the impact of microplastic pollution on freshwater fish and suggests the possible risk on fish sustainability as well as human health.

## CRediT authorship contribution statement

**Bini C. Das:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Aparna Ramanan P:** Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Satkar Sagar Gorakh:** Formal analysis. **Devika Pillai:** Funding acquisition, Resources, Project administration. **Rejish Kumar V.J:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data Availability

Data will be made available on request.

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